Geometric Intuition Behind Matrices in Practice

A Centroid-Based Approach to Binary Linear Classification

Technical Whitepaper

July 7, 2026

Abstract

This paper demonstrates the practical and geometric application of vector and matrix operations in the domain of credit scoring. By shifting focus from dry algebraic formulas to spatial intuition, we construct a simple yet mathematically rigorous binary linear classifier utilizing mass centroids, vector subtraction, and orthogonal projections.

1 Spatial Representation & Setup

Let every loan applicant be represented as a feature vector $\mathbf{x}$ within a bounded three-dimensional vector space $\mathbb{R}^3$, where the coordinates are defined by normalized parameters:

- x_1: Credit Score (CS)
- x_2: Work History (WH)
- x_3: Current Income (CI)

Based on historical data, applicants are partitioned into two disjoint subsets:

$$X_+ : \text{"Good" clients who successfully repaid their loans (Green cluster).}$$
$$X_- : \text{"Bad" clients who defaulted on their obligations (Red cluster).}$$

To eliminate individual noise and capture systemic group behavior, we calculate the geometric centers of mass (**centroids**) for both groups using their arithmetic means:

$$\mu_+ = \frac{1}{N_+} \sum_{\mathbf{x} \in X_+} \mathbf{x} \quad \text{and} \quad \mu_- = \frac{1}{N_-} \sum_{\mathbf{x} \in X_-} \mathbf{x} \tag{1}$$

Where N_+ and N_- represent the total number of samples in each respective class.

2 The Grading Vector & Dimension Reduction

To identify the axis of maximum classification variance, we construct a grading vector $\mathbf{w}$ by subtracting the bad centroid from the good centroid:

$$\mathbf{w} = \mu_+ - \mu_- \tag{2}$$

Algebraic Insight: Coordinate-wise subtraction filters out non-discriminative noise. If a specific metric shares uniform values across both groups, its difference yields 0, automatically excluding it from the scoring priority. The vector $\mathbf{w}$ effectively serves as a customized metric scale (ruler) for our task.

3 The Boundary Point & Invariant Mapping

The physical point of absolute equilibrium between the two historical clusters in our multi-dimensional space is the spatial midpoint $\mathbf{x}_0$:

$$\mathbf{x}_0 = \frac{\mu_+ + \mu_-}{2} \tag{3}$$

To establish a one-dimensional threshold, we orthogonally project this physical midpoint onto our grading vector $\mathbf{w}$ using the vector dot product. This defines our scalar threshold b:

$$b = \mathbf{w} \cdot \mathbf{x}_0 \tag{4}$$

The Invariant Core Principle: The transformation of data onto the grading vector compresses dimensions for computational efficiency but leaves the core geometric identities completely untouched. An applicant's position within the spatial volume remains structurally consistent whether evaluated in the multi-dimensional space or on the vector line. The point of equilibrium (the point of no return) preserves its boundary role across all dimensions.

4 The Operational Decision Rule

When evaluating a new applicant vector $\mathbf{x}_{\text{new}}$, the system projects their metrics directly onto the grading axis via the dot product:

$$\text{Score} = \mathbf{w} \cdot \mathbf{x}_{\text{new}} = w_1 x_1 + w_2 x_2 + w_3 x_3 \tag{5}$$

Geometrically, this classification boundary forms a flat 2D hyperplane in our 3D space, passing through $\mathbf{x}_0$ and perpendicular to $\mathbf{w}$. The system applies the following rule:

1. **If Score** $\geq b$: Approved (Safe Zone).
2. **If Score** $< b$: Rejected (Default Risk Zone).

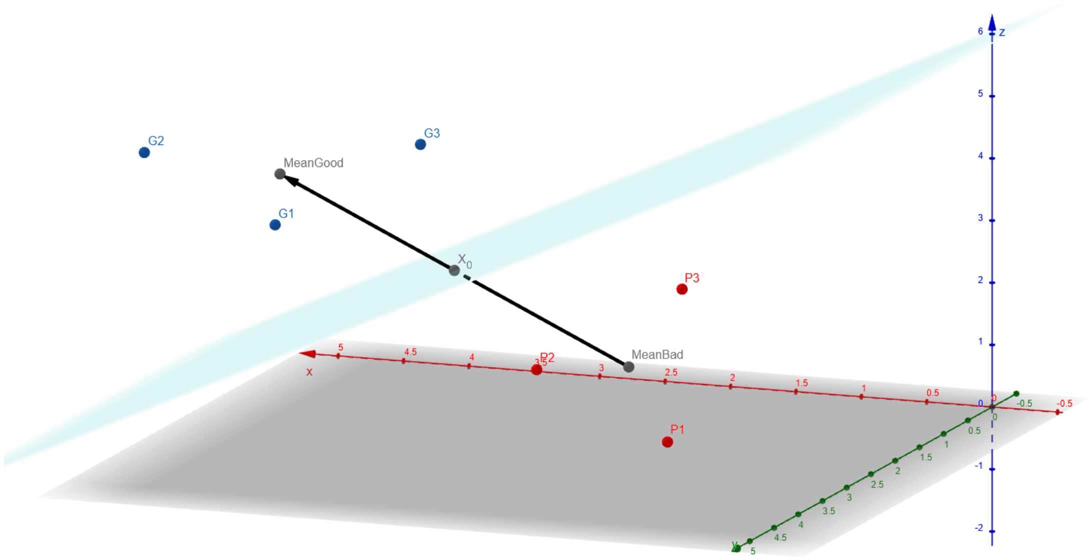

Figure 1: 3D Visualization of applicant clusters, the grading vector $\mathbf{w}$, and the separating hyperplane passing through the equilibrium point $\mathbf{x}_0$.

Figure 2: It's my very first project here, so I would highly appreciate any feedback, thanks for reading!